

# Penalized score selection Margin Preservation under Bounded Score Perturbations

Research Architecture Runtime  
operator/penalized-score-selection

2026-07-29

## Abstract

`penalized-score-selection` selects a unique maximizer. Margin  $\gamma(s) > 2\varepsilon$  is necessary and sufficient for unique-max invariance under  $\|\delta\|_\infty \leq \varepsilon$ . Lean `LEAN_FULL` (reduction to `Argmax.Margin`).

**This theorem is:** Reduction

## 1 Problem

The `penalized-score-selection` operator returns the unique maximizer index  $i^*$  of scores  $s \in \mathbb{R}^m$  ( $m \geq 2$ ), up to the operator's definitional presentation (heap top, tournament winner, masked max, etc.).

## 2 Stability notion

Perturbations:  $\|\delta\|_\infty \leq \varepsilon$ . Let  $\gamma(s) = s_{i^*} - \max_{j \neq i^*} s_j$ . Stability:  $\gamma(s) > 2\varepsilon$  implies  $i^*$  remains the unique maximizer of  $s + \delta$ . Sharpness:  $\gamma(s) \leq 2\varepsilon$  admits an adversary destroying unique maximality.

## 3 Definitions

**Definition 1** (Unique maximizer).  $i^*$  uniquely maximizes  $s$  if  $s_j \leq s_{i^*}$  for all  $j$  and  $s_j = s_{i^*} \Rightarrow j = i^*$ .

**Definition 2** (Margin).  $\gamma(s) = s_{i^*} - \max_{j \neq i^*} s_j$ .

## 4 Theorem

**Theorem 3** (Margin invariance). *If  $i^*$  uniquely maximizes  $s$  and  $s_{i^*} - s_j > 2\varepsilon$  for all  $j \neq i^*$ , then for every  $\|\delta\|_\infty \leq \varepsilon$ ,  $i^*$  uniquely maximizes  $s + \delta$ .*

**Theorem 4** (Sharpness). *If some  $j \neq i^*$  has  $s_{i^*} - s_j \leq 2\varepsilon$ , an admissible  $\delta$  destroys unique maximality.*

## 5 Intuition

Depressing the winner by  $\varepsilon$  and elevating a rival by  $\varepsilon$  consumes  $2\varepsilon$  of margin. Strict margin above  $2\varepsilon$  leaves the winner unique; otherwise a two-point adversary forces a tie or loss.

## 6 Examples

**Example 5.** Scores  $(7, 5, 4)$ ,  $i^* = 0$ ,  $\gamma = 2$ . For  $\varepsilon = 0.8$ ,  $2\varepsilon = 1.6 < 2$ , so the winner is stable. For  $\varepsilon = 1$ ,  $2\varepsilon = 2 \geq \gamma$ ;  $\delta = (-1, +1, 0)$  yields a tie between the first two scores.

## 7 Proof

Let  $s' = s + \delta$  with  $\|\delta\|_\infty \leq \varepsilon$ . For  $j \neq i^*$ ,

$$s'_{i^*} - s'_j = (s_{i^*} - s_j) + (\delta_{i^*} - \delta_j) \geq (s_{i^*} - s_j) - 2\varepsilon > 0,$$

using  $s_{i^*} - s_j > 2\varepsilon$ . Thus  $s'_j < s'_{i^*}$  for all  $j \neq i^*$ , so  $i^*$  uniquely maximizes  $s'$  (Theorem 3).

For sharpness, pick  $j$  with  $s_{i^*} - s_j \leq 2\varepsilon$ , set  $\delta_{i^*} = -\varepsilon$ ,  $\delta_j = +\varepsilon$ , and 0 elsewhere. Then  $\|\delta\|_\infty \leq \varepsilon$  and  $s'_{i^*} - s'_j = \gamma_{ij} - 2\varepsilon \leq 0$ , so unique maximality fails (Theorem 4).

## 8 Formal statement

### Lean module

`Research.Operators.PenalizedScoreSelection.Margin`

### Source file

`lean/Research/Operators/PenalizedScoreSelection/Margin.lean`

### Theorems

- `penalized\_score\_selection\_margin\_invariance`
- `penalized\_score\_selection\_margin\_sharpness`

### Note

Definitional reduction of `Argmax.Margin`.

### Certificate

`lean/certificates/penalized-score-selection/penalized-score-selection-margin`

### Status

LEAN\_FULL on Mathlib  $\mathbb{R}$  (REAL\_MATHLIB)

## 9 Proof dependencies

- `Research.Operators.Argmax.Margin` (`margin_invariance`, `margin_sharpness`).
- $\ell_\infty$  gap shrinkage by at most  $2\varepsilon$ .
- No other operator theorems required.

## 10 Consequences

Any unique-max presentation of `penalized-score-selection` inherits Argmax margin stability.  
Related: `argmax`, `tournament-winner`, `greedy-maximum-selection`.